What is it not?

To all the sci-fi fans out there, lets cut the fantasies and get down to the basics. We are going to tell you, in simple terms, what are the hard limits of Nanotechnology are and why some fantasies are simply not feasible.

Not Alchemy

Nanotechnology is not alchemy; it cannot turn lead into gold. That would require changing the structure inside atoms. Nanotechnology can only build objects using atoms. Rearranging atoms does change the properties of materials. The only difference between diamond and charcoal is the arrangement of its carbon atoms. But that does not mean we can make any material infinitely strong. True, we can make rockets of atomically perfect steel with no hidden cracks to give way over time. We could even build that rocket of diamond. But even diamond can be broken; it is only as strong as its chemical bonds. Nanotechnology does not give us control over chemistry.

Raw Materials

Nanotechnology could build diamond as easily as charcoal. But to do either, it would still need carbon atoms. The same principle would apply to anything we want to build. Although we still have a wealth of mineral resources on Earth, the scale of building with nanotechnology could conceivably outstrip that supply. In the long term, an endless supply of building materials exists in outer space, particularly in asteroids. But in the short term, scarcity of materials could be a limit on nanotechnology designs.

Random radiation

Natural radiation (fast moving particles that can penetrate objects) may not be our everyday concern. But it does cause damage at the atomic scale by disturbing the arrangement of atoms. For example, the rearrangement of atoms in human cells by radiation commonly results in cancer. If a nanotechnology machine is only 50 atoms across, radiation could significantly damage it. Even large scale machines built using nanotechnology might be in danger if they rely on a perfect arrangement of atoms to operate correctly.

Atomic magnets

Although it makes a great visual image, atoms cannot really be manipulated like building blocks. Conditions at the atomic level are much different than at human scale. Atoms behave somewhat like magnets. When they are brought too close together, overlap repulsion forces push them apart; at a distance, Van der